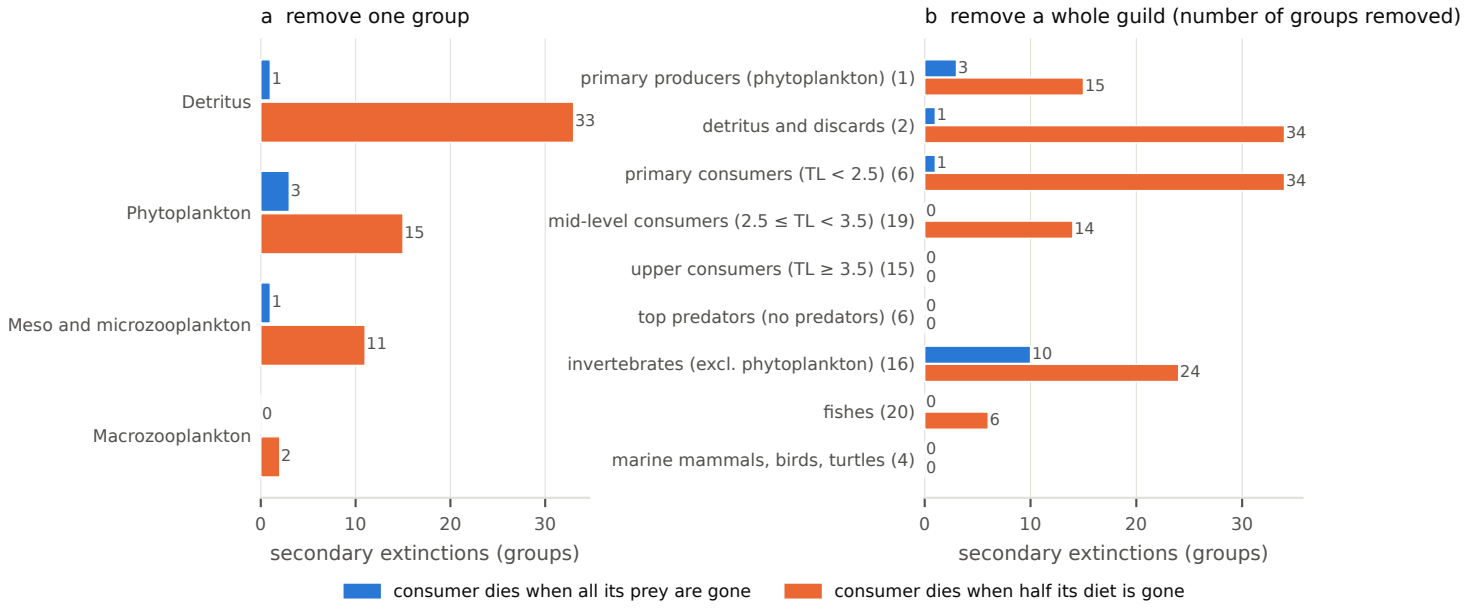
a remove one group
Detritus
Phytoplankton
Meso and microzooplankton
Macrozooplankton
1
33
3
15
1
11
0
2
0
10
20
30
secondary extinctions (groups)
b remove a whole guild (number of groups removed)
primary producers (phytoplankton) (1)
detritus and discards (2)
primary consumers (TL < 2.5) (6)
mid-level consumers (2.5 ≤ TL < 3.5) (19)
upper consumers (TL ≥ 3.5) (15)
top predators (no predators) (6)
invertebrates (excl. phytoplankton) (16)
fishes (20)
marine mammals, birds, turtles (4)
3
15
1
34
1
34
0
14
0
0
0
0
10
24
0
6
0
0
0
10
20
30
secondary extinctions (groups)
consumer dies when all its prey are gone
consumer dies when half its diet is gone